1.A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Step	What You Are Checking For
1. Check the Power Adapter and Power Outlet	Ensure the charger is working properly and the wall socket is supplying power.
2. Inspect the Charging Cable and Port	Check for a damaged cable, loose connector, or dust and damage in the charging port.
3. Check the Battery	If the battery is removable, remove and reconnect it or try powering on with only the charger connected.
4. Perform a Power Reset	Hold the power button for 15–20 seconds to discharge residual power and reset the laptop.
5. Check the Power Indicator LEDs	Observe whether the charging LED or power LED lights up. If there is no LED activity, the issue may be with the charger, battery, DC jack, or motherboard.

2.You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.

Step	What to Check
1. Check the Monitor Power	Ensure the monitor is turned on and the power LED is glowing.
2. Check the Display Cable	Verify that the HDMI, VGA, DVI, or DisplayPort cable is securely connected to both the monitor and the PC.
3. Check the Input Source	Make sure the monitor is set to the correct input source (HDMI, VGA, DisplayPort, etc.).
4. Restart the Computer	Restart the PC to see if the display appears during boot.
5. Test with Another Monitor or Cable	Replace the display cable or connect another monitor to determine whether the monitor or cable is faulty.
6. Check the Graphics Card	Ensure the GPU is properly seated and, if required, that its power connectors are securely attached.
7. Check the RAM	Reseat the RAM modules, as loose or faulty RAM can prevent the display from appearing.
8. Confirm the Fix	After correcting the issue, restart the PC and verify that the monitor displays the desktop normally.

3.A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

Step	Troubleshooting Action	Purpose / What You Are Checking
1	Check the mouse power and battery.	Ensure the mouse is switched on and the batteries are not dead. Replace the batteries if necessary.
2	Check the wireless USB receiver.	Verify that the USB receiver is firmly connected and not damaged.
3	Test the USB receiver in another USB port.	If the mouse works in another port, the original USB port may be faulty.
4	Test the mouse on another computer.	If the mouse works on another system, the mouse is functioning correctly and the issue is with the original computer.
5	Check the system settings and drivers.	Open **Device Manager** and confirm that the mouse is detected, enabled, and the driver is installed correctly. Restart the system if required.

4.Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hint: Think about fans, dust, and thermal paste.

Possible Hardware Cause	How to Check
1. Faulty or Non-Working Cooling Fan	Check whether the CPU fan and case fans are spinning properly. Replace any fan that is not working or making unusual noise.
2. Dust Build-up Inside the PC	Open the PC case and inspect the CPU heatsink, cooling fans, and air vents for dust. Clean them carefully using compressed air or a soft brush.
3. Dried or Improper Thermal Paste	Remove the CPU heatsink and inspect the thermal paste. If it is dry or uneven, clean the old paste and apply a fresh layer before reinstalling the heatsink.